

Clinical Risk Assessment Report: Patient 88

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 39.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.77x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.43 [Bottom 1%] (+5.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 48.0 [Bottom 7%] (+5.0%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.15 [Top 6%] (+3.6%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.66 [Top 24%] (+2.7%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: 2.06 [Top 2%] (+1.8%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.4 [Top 4%] (+1.5%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.23 [Bottom 8%] (+1.3%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.17 [Top 12%] (-3.0%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)

